

Tutorial 2: Exploratory Data Analysis & Distribution Visualization

Distribution shape, the canonical visual signatures, Exploratory Data Analysis (EDA), Box plots, histograms, kernel density estimates.

Learning outcomes:

Interpret data distribution using skewness, kurtosis, and visualization techniques.

Perform Exploratory Data Analysis (EDA) using box plots, histograms, and kernel density estimates (KDE).

1. A data analyst collected the following customer purchase amounts (₹):

120, 135, 140, 145, 150, 152, 155, 158, 160, 165, 170, 175, 180, 190, 300

Perform an exploratory data analysis by:

a) Finding the five-number summary (Minimum, 1, Median, 3, Maximum).

b) Calculate the IQR and determine whether the dataset contains any outliers using the $1.5 \times \text{IQR}$ rule.

c) Based on your findings, comment on the likely shape of the distribution.

Solution:

(a) Five-Number Summary

Data are already in ascending order.

- a) **Minimum = 120**
- b) **$Q_1 = 145$**
- c) **Median (Q_2) = 158**
- d) **$Q_3 = 175$**
- e) **Maximum = 300**

Five-Number Summary

Statistic	Value (₹)
Minimum	120
First Quartile (Q_1)	145
Median (Q_2)	158
Third Quartile (Q_3)	175
Maximum	300

(b) Calculate the IQR and Identify Outliers

Interquartile Range (IQR)

$$\begin{aligned}\text{IQR} &= Q_3 - Q_1 \\ &= 175 - 145 = 30\end{aligned}$$

$$\boxed{\text{IQR} = 30}$$

Calculate the Outlier Limits

Lower Fence:

$$Q_1 - 1.5(IQR) = 145 - 1.5(30) = 145 - 45 = 100$$

Upper Fence:

$$Q_3 + 1.5(IQR) = 175 + 45 = 220$$

Outlier limits:

- a) **Lower Limit = 100**
- b) **Upper Limit = 220**

Since **300 > 220**, **300 is an outlier**.

$\text{Outlier} = 300$

(c) Comment on the Shape of the Distribution

- a) The dataset contains **one high-value outlier (300)**.
- b) Most purchase amounts are clustered between **120 and 190**.
- c) The distribution has a **long tail toward higher values**.

The distribution is positively (right) skewed due to the high-value outlier (300).

2. The following summary statistics are obtained for the monthly electricity consumption (kWh) of households:

• Minimum = 180

• Q1 = 240

• Median = 290

• Q3 = 360

• Maximum = 620

a) Calculate the IQR.

b) Determine the lower and upper fences for outlier detection.

c) State whether the maximum value is an outlier.

d) Sketch the corresponding box plot and comment on the distribution.

Solution:

(a) Calculate the IQR

$$\begin{aligned} \text{IQR} &= Q_3 - Q_1 \\ &= 360 - 240 = 120 \end{aligned}$$

$\text{IQR} = 120$

(b) Determine the Lower and Upper Fences

Lower Fence:

$$\begin{aligned}Q_1 - 1.5 \times \text{IQR} \\= 240 - 1.5(120) = 240 - 180 = 60\end{aligned}$$

Upper Fence:

$$\begin{aligned}Q_3 + 1.5 \times \text{IQR} \\= 360 + 180 = 540\end{aligned}$$

Answer:

- a) **Lower Fence = 60**
- b) **Upper Fence = 540**

(c) State Whether the Maximum Value is an Outlier

Maximum value = **620**

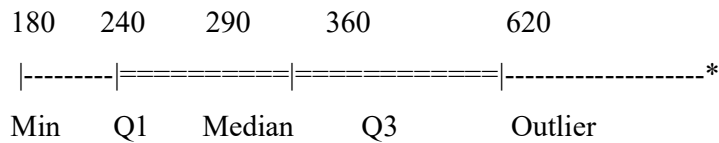
Since

$$620 > 540$$

Yes, 620 is an outlier.

(d) Sketch the Box Plot and Comment on the Distribution

Box Plot (Approximate)



Comment:

- The value **620** is a high-value outlier.
- The distribution is **positively (right) skewed**.

3. The following frequency distribution represents the ages of customers visiting a retail store.

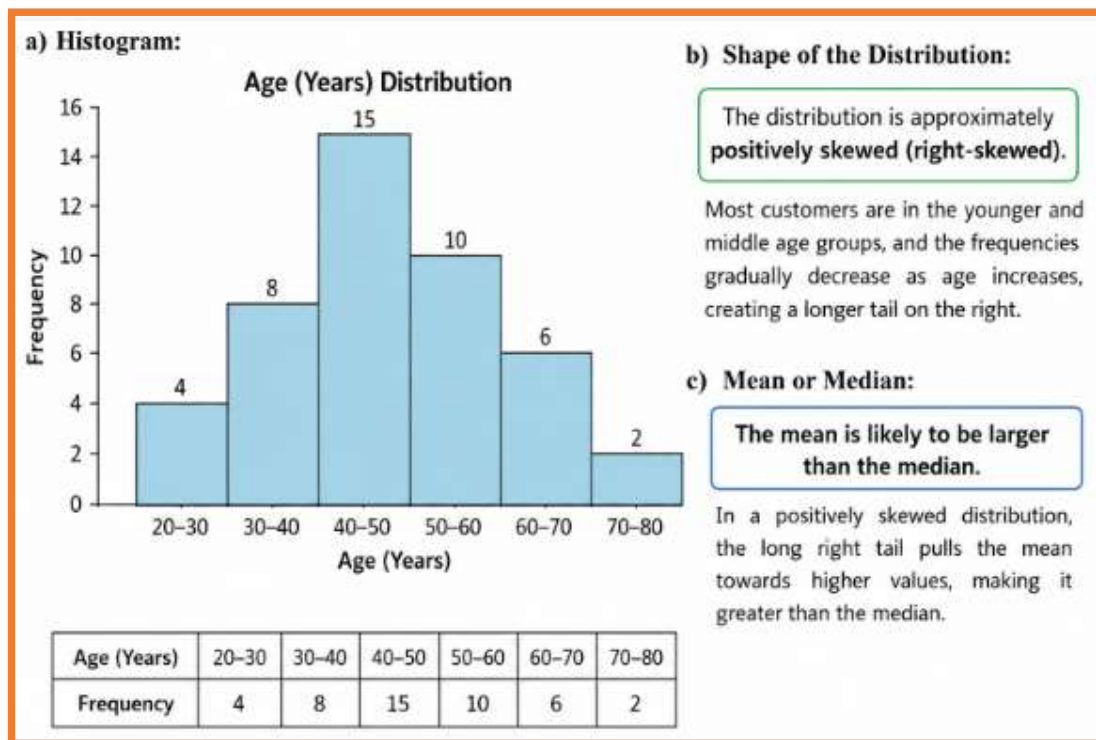
Age (Years)	Frequency
20–30	4
30–40	8
40–50	15
50–60	10
60–70	6
70–80	2

a) Draw a histogram for the given data.

b) Identify whether the distribution is approximately symmetric, positively skewed, or negatively skewed.

c) State whether the mean or median is likely to be larger and justify your answer.

Solution:



4. A dataset has the following descriptive statistics:

- Mean = 68

- Median = 61

- Standard Deviation = 14

- Pearson's coefficient of skewness:

$$Sk = \frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

Also, the kurtosis coefficient is 4.8.

a) Calculate the coefficient of skewness.

b) Classify the distribution as positively skewed, negatively skewed, or symmetric.

c) Identify whether the distribution is platykurtic, mesokurtic, or leptokurtic.

Solution:

(a) Calculate the Coefficient of Skewness

$$\begin{aligned} Sk &= \frac{3(68 - 61)}{14} \\ &= \frac{3 \times 7}{14} = \frac{21}{14} = 1.5 \end{aligned}$$

$$\text{Coefficient of Skewness} = 1.5$$

(b) Classify the Distribution

Since the coefficient of skewness is **positive** ($1.5 > 0$),

The distribution is positively (right) skewed.

(c) Classify the Distribution Based on Kurtosis

For kurtosis:

- a) $< 3 \rightarrow$ Platykurtic
- b) $= 3 \rightarrow$ Mesokurtic
- c) $> 3 \rightarrow$ Leptokurtic

Given:

$$\text{Kurtosis} = 4.8 > 3$$

The distribution is leptokurtic.

5. A data scientist records the following response times (ms):

82, 85, 86, 88, 90, 91, 92, 93, 95, 96, 98, 100, 102, 150

a) Compute the five-number summary.

b) Explain whether a histogram or a kernel density estimate (KDE) would better reveal the distribution pattern, considering the data.

c) Identify whether the dataset is likely to be right-skewed, left-skewed, or symmetric based on the given values.

Solution:

(a) Compute the Five-Number Summary

The data are already arranged in ascending order.

- a) **Minimum = 82**
- b) **$Q_1 = 88$**
- c) **Median = $(92 + 93)/2 = 92.5$**
- d) **$Q_3 = 99$**
- e) **Maximum = 150**

Five-Number Summary

Statistic	Value (ms)
Minimum	82
Q_1	88
Median	92.5
Q_3	99
Maximum	150

(b) Histogram or KDE?

Kernel Density Estimate (KDE) is better because it provides a smooth view of the distribution and clearly highlights the right tail caused by the value **150**. A histogram may depend on the choice of bin width.

(c) Identify the Shape of the Distribution

The value **150** is much larger than the remaining observations, creating a long tail on the right.

The dataset is positively (right) skewed.

VIVA-VOCE Questions:

1. What does positive skewness indicate?

Positive skewness indicates that the distribution has a **long tail on the right**, with most values concentrated on the left.

2. What is kurtosis?

Kurtosis is a measure of the **peakedness and tail heaviness** of a data distribution.

3. What is the purpose of Exploratory Data Analysis (EDA)?

EDA is used to **summarize, visualize, and understand** the main characteristics of a dataset before detailed analysis.

4. What information can a box plot provide?

A box plot shows the **five-number summary**, spread of the data, and helps identify **outliers**.

5. How is a histogram different from a Kernel Density Estimate (KDE)?

A **histogram** displays data using frequency bars, while a **KDE** shows a smooth curve representing the data distribution.
